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5G

“You Need To Take That L-SHAPED PCB And Figure Out How To Put MILLIMETRE WAVE Into It”

There is a lot of talk about 5G, its benefits and applications. For engineers planning to start playing around with 5G, it's important to get a bird's eyeview of the challenges in designing a mobile device using this technology. Jim Cathey, Larry Paulson and Peter Carson from Qualcomm explain the basics to EBY's Dilin Anand

Q. What design changes are required to make smartphones millimetre-wave-capable?

A. If you take any smartphone, it has an L-shaped PCB at one side and rest of the space is carved out for a battery. You need to take that L-shaped PCB (which is already packed to the brim) and figure out how to put millimetre wave into it. You cannot use the existing antenna setup for LTE here, because the antenna array and other aspects are completely different from what is needed for millimetre wave. Overall, there are three items that need to be balanced: power, coverage and size.

Q. Could you explain power design challenges?

A. Power problems are more of a downlink problem. The gain in your amplifier decides the

amount of power you transmit back to the base station. Power-amplified antenna and the choice of the antenna element have an important impact here. Certain antenna designs cause the radiation to go forward and narrow instead of widening and shortening. The power amplifier drives each one. So if you have four antennae and power amplifiers, you can add them together so that these direct to perform beamforming.

Q. What's the best way to maximise gain here?

A. You can go for an extremely low-power amplifier, and add four 4dB amplifiers together to get 16dB gain. Remember this is going to be a logarithmic addition for dB here, so it's an order of magnitude higher for each. Once this is done, you can use polarisation to get an additional 3dB of antenna gain. The most severe power efficiency impacts will be at the downlink due to multi-gigabit data streams, data processing and related aspects.

Q. What's the biggest benefit of using 802.11ad instead of 802.11ac?

A. 802.11ad gives much higher throughputs at the same power level. The aim here is to balance the power throughput and not just look at the power efficiency.

Q. Any final tip that designers need to keep in mind?

A. You may still end up grabbing the wrong end of the device and blocking the module. So you need to design a way out for this by having more than one module in these millimeter-wave devices.

Q. What will be the scenario ten years from now?

A. In the future, you will have less low-level decisions to make. At the same time, you could also have a lot of data on tap that can be easily accessed through a potentially augmented reality interface. Overall, this means that you will have more time to dwell on the things of your choice while everything else can be automated.

Q. Autonomous vehicles are starting to look a lot like gadgets. What's your take on a realistic scenario?

A. What will make autonomous driving actually work is a 100 per cent autonomous driving system. If you ban driver-occupied cars between 6am and 10pm in New York, London, Tokyo or any other major city for instance, and then smart grid the lighting system in the roads, the whole thing is going to work in an automated way. It is going to be an extremely efficient way. **EBY**